# Fair Allocation of Job Training under Uncertain Treatment Benefits

ICT 5143: AFGML Capstone | Problem 6

Akshay Wali (261090830018) Mohit Das (261090840018)

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**Testable hypothesis.** With candidates and capacity fixed, the lower-bound rule should select fewer policies that violate the true group floor than the point-estimate rule in small-group simulations.

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- **Viviano & Bradic (2024).** Fair policy targeting motivates examining group welfare separately from population value.
- **Our comparison.** Keep the same stump candidates and compare no floor, a point-estimate floor and a simultaneous lower confidence bound (LCB). Track true violations, infeasibility and zero group access in simulation.

Eight-paper comparison and restricted baseline details in `docs/related_work.md`.

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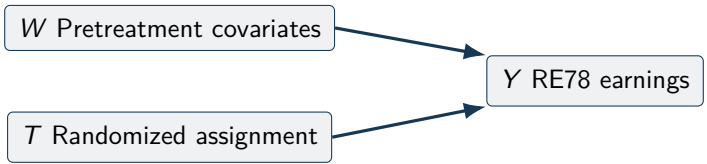
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**Pooled outer-test support**

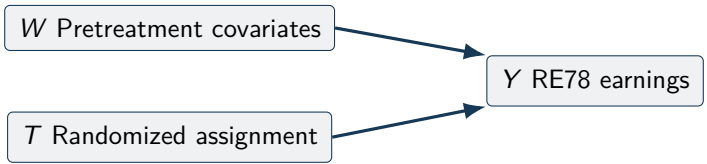
| Group | | Treated | Control |
|---|---|---|---|
| Black | | 156 | 215 |
| Hispanic | | 11 | 28 |
| Neither recorded | indicator | 18 | 17 |

The last two groups have fewer than 30 participants in either arm. Treat those arm-level results as exploratory.

# Graphical model and inference



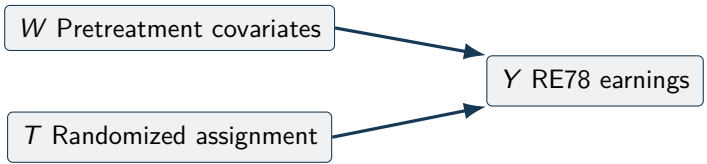
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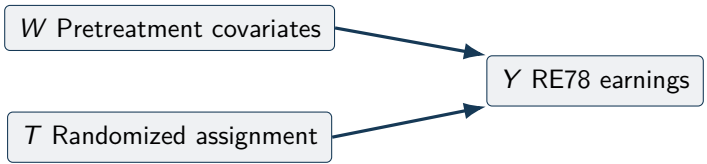
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- In the separate discrete fixture, VE and clique-tree inference both give $P(Y = 1 \mid T = 1, X = 1) = 0.70$.
- Six queries passed. Maximum discrepancy was $5.55 \times 10^{-17}$ at tolerance $10^{-8}$.

# Fairness criterion and proposed method

$$V(d) = \mathbb{E}[d(W)\{Y(1) - Y(0)\}], \qquad B_g(d) = \mathbb{E}[d(W)\{Y(1) - Y(0)\} \mid G = g].$$

- **Utility.** Maximize incremental earnings per eligible person at fixed capacity.

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- Race is used in constrained validation and ridge ranking. Final stumps use age if it is the chosen split. Race indicators are excluded from the final stump inputs.

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- Five seeds for simulations and refits. Paired intervals use 2,000 participant bootstrap draws.
- Evaluate 20%, 40% and 60% capacity, transformed earnings, clipping and simulated confounding.

**Protocol status.** Existing results are exploratory. Holm is specified for a future independent evaluation.

# Main result and uncertainty

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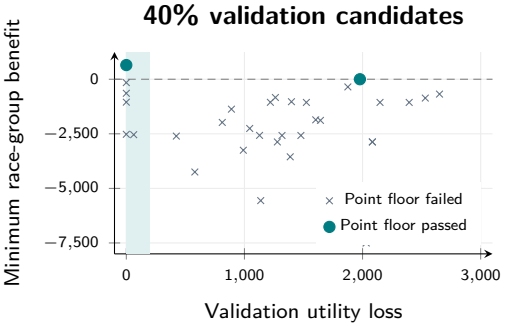
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**Matched validation utility.** Point estimate 1/5 folds. LCB 0/5. The paired held-out comparison with the LCB could not be estimated.




Shading marks the 200-unit tolerance. All 34 candidates failed the LCB floor.

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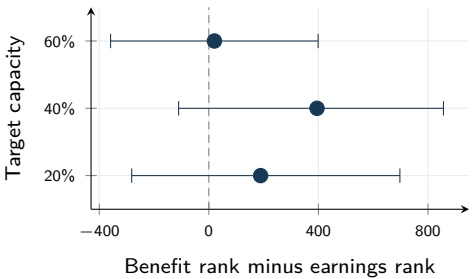
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**Ranking ablation**




Paired 95% intervals for earnings gain per eligible person. All include zero.

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- **Akshay Wali:** setup, experiment runs and clean reproduction. **Mohit Das:** individual contribution and peer cross-check pending confirmation.

# Backup: mandatory evidence and references

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**Selected references**


- Athey & Wager (2021), *Econometrica*. DOI 10.3982/ECTA15732.
- Kitagawa & Tetenov (2018), *Econometrica*. DOI 10.3982/ECTA13288.
- Viviano & Bradic (2024), *JASA*. DOI 10.1080/01621459.2022.2142591.
- Dehejia & Wahba (1999), *JASA*. NSW RE74 data documentation.

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Experimental ATE: 1,794.34 (95% interval [518.53, 3,132.21]).

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3. Use `results/logs/run_all.log` as the saved demonstration trace.